

Ruby (Yuhan) Qiu, EIT, CAPM Candidate

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HIGHLIGHTS OF QUALIFICATIONS

- **Embedded Systems & Motor Control:** Designed and implemented real-time motor control algorithms for autonomous robots, achieving 5% improved speed regulation, $\pm 30\%$ load variation tolerance, and reduced overshoot by 35% using PID control and ROS integration.
- **Energy & Reliability Analysis:** Assisted in monitoring and analyzing 50+ electrical and instrumentation systems, calculating MTBF, identifying failure trends, and supporting preventive maintenance programs that reduced unplanned downtime by 5%.
- **FPGA & Hardware Design:** Developed 16-channel FPGA signal processing pipelines for wearable ultrasound and SD-card host systems, achieving < 5 ms latency, 20% improved signal quality, and reliable data streaming at ~ 2 MB/s across multiple platforms.
- **Data Analysis & Visualization:** Skilled in Power BI, Excel, and MATLAB for processing energy, telematics, and biomedical data; prepared dashboards, KPIs, and reports supporting cross-functional decision-making and program optimization.
- **Cross-Functional Collaboration & Research:** Experienced in interdisciplinary collaboration with engineers, researchers, and clinicians; contributed to AI data annotation, embedded systems projects, and reliability analysis, ensuring 100% compliance with ethics, standards, and regulatory requirements.

EDUCATION

Bachelor of Applied Science in Electrical Engineering
University of British Columbia

Sep 2022 - May 2026
Vancouver, Canada

Certified Associate in Project Management®
Project Management Institute

Jan 2026 - Present
Vancouver, Canada

EXPERIENCE

Navigation & Embedded Systems Member
UBC Agrobot

Sep 2025 - Present
Vancouver, Canada

- **Motor Control Design:** Designed and tested real-time motor control logic for an autonomous field robot, improving speed regulation accuracy by 5% and maintaining stable operation under load variations of up to $\pm 30\%$.
- **ROS Integration:** Implemented ROS and micro-ROS interfaces enabling reliable motor command and feedback exchange at 100 Hz, reducing communication latency by $\sim 40\%$ compared to the initial prototype.
- **Hardware Interfacing:** Integrated motor drivers, quadrature encoders, and motor controllers to achieve closed-loop control with encoder resolution up to 1024 PPR and position error below 2%.
- **Control Algorithms:** Developed and tuned PID-based motor control algorithms, decreasing overshoot by 35% and settling time by 20% during start-stop and turning maneuvers.
- **Embedded Programming:** Implemented motor control and communication modules in C++ and Python, supporting real-time execution on embedded hardware with over 99% uptime during field and bench testing.

AI Research Assistant
Uterine Health Research Lab, UBC Faculty of Medicine

Jun 2025 - Nov 2025
Vancouver, Canada

- **Medical Data Annotation:** Annotated and validated 200+ ultrasound imaging datasets for AI model development, achieving 100% compliance with de-identification, ethics approval, and licensing requirements.
- **Data Quality Assessment:** Evaluated dataset quality by reviewing 50+ medical publications and associated metadata, improving dataset consistency and usability for model training.
- **Standards & Compliance:** Enforced data governance standards for sensitive medical imaging data, reducing rework and compliance review time by $\sim 30\%$.
- **Research Documentation:** Documented dataset attributes, limitations, and quality metrics in a structured spreadsheet used by a 6-person research team for downstream development.
- **Cross-Functional Collaboration:** Collaborated with clinicians and AI researchers to define data selection and quality criteria, directly supporting successful preparation of datasets for model prototyping and validation.

Electrical Engineering Intern - Telematics
Jimi Group Ltd.

May 2024 - Aug 2024
Huizhou, China

- **Embedded Systems:** Assisted in the development and testing of vehicle telematics hardware and microcontroller-based systems, supporting 10+ GPS tracking devices and 5 CAN bus modules for fleet applications.

- **Testing & Validation:** Performed signal validation and electrical testing (voltage, current, signal integrity) using oscilloscopes, multimeters, and logic analyzers, achieving 99% test pass rate across 50+ device prototypes.
- **Firmware Integration:** Supported firmware integration and debugging for microcontroller-based telematics devices, improving sensor data processing accuracy by 15% and reducing power consumption by ~8%.
- **Data Analysis:** Analyzed telematics logs (location, speed, power status, fault codes) from 200+ test runs, identifying performance issues and supporting reliability improvements that reduced fault reports by 20%.
- **Hardware Development:** Assisted engineers with hardware bring-up, wiring, and PCB validation for 15 prototype boards, including connector verification and power supply testing, ensuring compliance with design specifications.
- **Documentation:** Prepared 6+ technical reports and test summaries, including procedures, results, and recommendations, supporting cross-functional team review and product validation.

Reliability Engineer Intern - E&I Yongxin Factory Co. Ltd.

May 2023 - Aug 2023
Quanzhou, China

- **Equipment Reliability:** Assisted in monitoring and analyzing performance data of 50+ electrical and instrumentation systems, identifying recurring faults and recommending preventive maintenance actions.
- **Data Analysis:** Collected and processed historical operational data using Power BI and Excel to calculate Mean Time Between Failure (MTBF) and identify failure trends for critical E&I equipment.
- **Preventive Maintenance Support:** Assisted in developing and updating preventive maintenance schedules for motors, sensors, PLCs, and control panels, contributing to an estimated 5% reduction in unplanned downtime.
- **Failure Investigation:** Participated in root cause analysis (RCA) for 10+ minor electrical and instrumentation failures, documenting findings and corrective actions for engineering review.
- **Standards & Compliance:** Supported verification of E&I systems against local electrical safety standards and factory procedures, improving audit readiness and regulatory compliance.
- **Reporting & Documentation:** Prepared 5+ technical reports summarizing reliability metrics, equipment health assessments, and maintenance recommendations, facilitating cross-team decision-making.

PROJECT EXPERIENCE

Miniature Ultrasound Transducer for Wearable Medical Imaging

2025

- **FPGA Design & Signal Processing:** Developed a 16-channel preprocessing pipeline handling 1 MS/s per channel, including synchronized acquisition, filtering, envelope detection, and decimation, enabling real-time wireless transmission with <5 ms latency.
- **Control Interface Implementation:** Designed and implemented FPGA-AFE interfaces for programmable excitation timing, channel selection, VGA gain control, and Doppler support, improving signal quality by ~20% and supporting multiple scanning modes.
- **Modular Hardware Development:** Contributed to a scalable PCB architecture separating FPGA SoM, high-voltage pulser, and 16-channel AFE, reducing debugging time by 30% and supporting easy system upgrades for future 32-channel expansion.

FPGA SD-card Reader

2025

- **FPGA Development:** Implemented an SD-card host in SystemVerilog supporting SDv1, SDv2, SDHC cards, and FAT16/FAT32 file systems, enabling read/write operations on cards up to 64 GB.
- **Data Handling & File Access:** Built sector-level read logic and file-system parsing to locate and stream over 500 non-contiguous files, achieving continuous data throughput of ~2 MB/s with no data corruption.
- **Validation & Testing:** Performed full RTL simulation and on-board verification, reducing read/write errors by over 10% and ensuring reliable operation across 5 different SD-card models.

Cardio Health Monitor

2024

- **Embedded System Development:** Implemented a real-time heart rate monitoring system in C, providing live visualization of heart rate variations from 50–180 BPM on an oscilloscope and health feedback on an LCD display.
- **Sensor Design & Integration:** Developed a PPG sensor with analog front-end (AFE) and ADC for amplified pulse detection, achieving ± 2 BPM measurement accuracy under varying heart rate conditions.
- **Testing & Verification:** Validated signal integrity and timing using laboratory instrumentation over 50+ test cycles, ensuring reliable readings and stable system performance.

TECHNICAL SKILLS

- **Programming / Software Skills:** C, C++, Python, Verilog/SystemVerilog, Vivado, ModelSim, Quartus, MATLAB, Power BI/Tableau, VBA
- **Embedded & Systems:** MCU and FPGA-based systems (STM32, ESP32, DE1-SoC, Xilinx), peripheral communication (ADC, UART, SPI, I2C), real-time systems, system-level debugging and validation
- **Laboratory & Testing:** Waveform generator, oscilloscope, multimeter, soldering, power supply